GlucoTwin architecture

a digital twin that forecasts post-meal glucose spikes 2 hours ahead

1 Open data

CGMacros bio.csv

45 adults: healthy, prediabetes, type 2 diabetes
Age, sex, BMI, HbA1c, fasting glucose and insulin, lipid panel

CGMacros minute data

Dexcom G6 + Libre Pro CGM
Fitbit heart rate, METs, activity calories
Logged meals: carbs, protein, fat, fiber (10 days each)

Synthetic EHR fields

Flagged in every record: diagnoses, medications, TCF7L2 genotype

2 Two streams

Static stream: EHR

One FHIR R4 bundle per patient (ABDM-compatible format)

Patient, Observation (LOINC), Condition (SNOMED, ICD-10), MedicationStatement

Parsed back into ~16 static features

Dynamic stream: wearables

Causal 5-minute grid: the row at t uses data up to t only

Original CGM readings recovered (CGMacros interpolated to 1 min)
Last reading held <=10 min; longer gaps stay missing
HR and METs averaged
Quality report per patient

Leakage tests on raw data

3 Fusion layer

Feature engineering

Glucose now, lags, trend, acceleration, 1-6 h stats, EWMA, 24 h baseline, time above 180

Carbs on board and absorption rate (gamma absorption model)
Meal macros in time windows, time since meal, fat on board

Activity: METs windows, active minutes, heart rate relative to resting

Time of day, missingness flags

EHR features: HbA1c, HOMA-IR, TG/HDL, BMI, age, diagnosis

Plan inputs (what-if): planned carbs, planned activity hidden at random in training

93 features per step, plus 6 h sequences for the network

4 Twin engine

LightGBM engine

Glucose at 30/60/90/120 min
Spike and hypo classifiers
Monotone constraints: the safe what-if engine

P-MRGF neural engine

TCN on 6 h CGM, FiLM from EHR
Gated residuals: meals, activity, heart rate
24-step trajectory + risk head

Honest ensemble + band

Per-horizon weights fixed on development patients; 80% band

Personalization

Per-patient residual model from the first days

5 Doctor-facing

What-if simulator

Swap the meal, change portion, add a walk and compare curves
Indian meal library (values to verify vs IFCT 2017)

Explanations

TreeSHAP contributions grouped into plain-language drivers

Doctor dashboard (Streamlit)

Forecast, what-if, health record, clinic overview ranked by risk, model evidence

REST API (FastAPI)

/forecast /whatif /ehr
ready for EHR/app integration

Validation 9 locked test patients scored once | 5-fold patient CV on 36 development patients | purged personalization | baselines | RMSE, R2 of change, Clarke grid, AUROC, AUPRC, Brier, false alerts/day